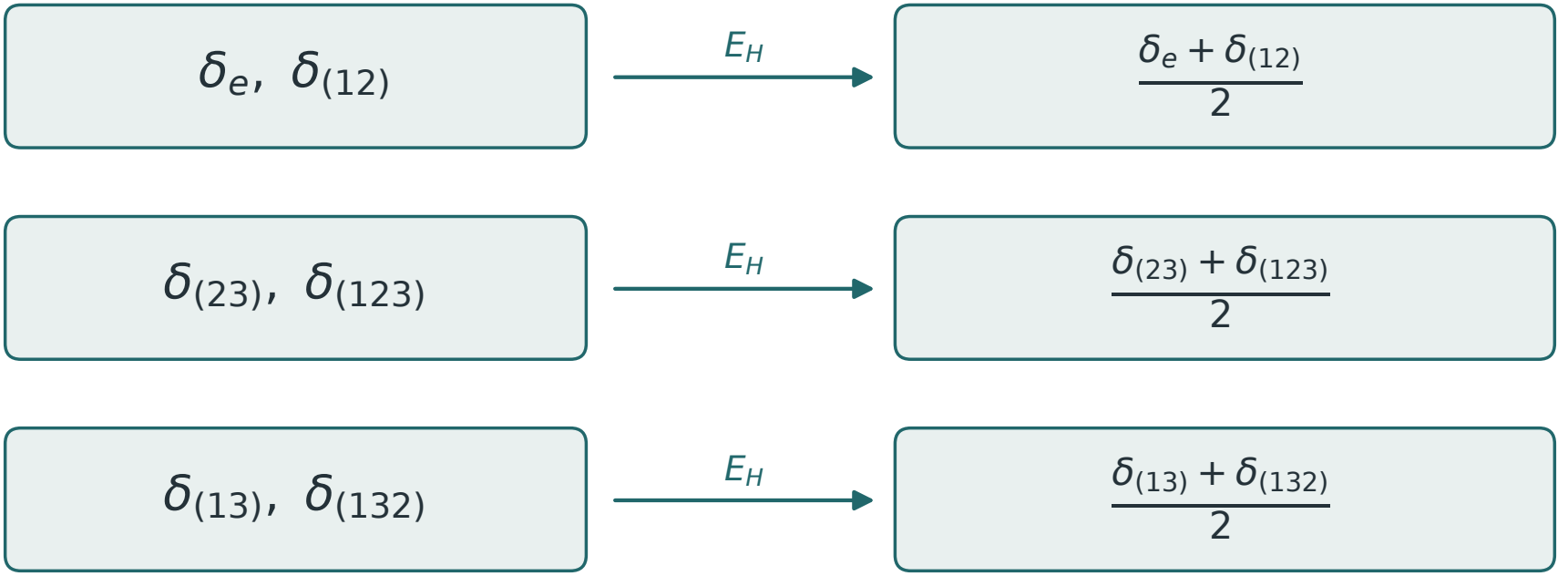
An average turns orbits into invariant coordinates

WRITTEN PROOF / left regular action on Q[S₃]



$$H = \langle (12) \rangle, \quad E_H^2 = E_H, \quad \operatorname{rank}(E_H) = 6/2 = 3$$